



CSE 3433- SOFTWARE ARCHITECTURE

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Topic: Event-Driven Job Recruitment & Application
Processing

System (Event-Driven Architecture)

Group: Group 12

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1. PROJECT PROPOSAL

1.1 SELECTED ARCHITECTURE STYLE

To efficiently manage communication and processing between different components of the system, the Event-Driven Job Recruitment & Application Processing System adopts an Event-Driven Architecture (EDA) style. This architecture allows system components to communicate through events that are generated whenever an important action occurs within the system. In this approach, different services operate independently but respond to specific events triggered by user actions.

In this system, user such as employers and job applicants interacts with the platform through the user interface. When employer posts a job vacancy or when an applicant submits a job application, an event is triggered and sent to the event processing mechanism.

Using Event-Driven Architecture allows the system to achieve loose coupling between components, improving scalability, flexibility, and maintainability when handling multiple recruitment activities. Therefore, the proposed system applies Event-Driven Architecture to support efficient recruitment processing, automated workflows, and scalable system performance. or instance, once an application is submitted, a screening service may automatically evaluate the resume, while a notification service sends confirmation messages to the applicant. This parallel processing significantly reduces delays and improves overall system efficiency.

1.2 PROBLEM DESCRIPTION

Many organizations receive many job applications when the advertise job vacancies. However, the recruitment process is often handled using traditional or centralized systems that process tasks sequentially. This approach can cause delays in processing applications, difficulties in tracking applicant status, and inefficiencies when managing multiple recruitment activities at the same time. However, in many cases, the recruitment process is still handled using traditional or centralized systems that process tasks sequentially.

Besides, applicants may not receive timely updates regarding their application status, while recruiters may struggle to manage tasks such as screening resumes, shortlisting candidates and sending notifications efficiently. Without an automated and responsive system, the recruitment process can become slow and difficult to manage. Therefore, an event-driven recruitment system is needed to automate workflows and improve the efficiency of handling job applications. f one part of the system becomes slow or fails, it can disrupt the entire recruitment workflow. This lack of responsiveness makes it difficult for organizations to adapt quickly to changing hiring needs or unexpected increases in application volume.

1.3 SYSTEM OBJECTIVES

1. To develop a centralized system that allows employers to post job vacancies and applicants to view available job opportunities efficiently.
2. To implement an application management mechanism that supports job application submission, tracking, and status updates.
3. To design the system using an event-driven approach, ensuring loose coupling and efficient communication between system components.
4. To provide a notification feature that keeps applicants informed about their application status and recruitment progress.
5. To improve recruitment efficiency by automating application processing and managing recruitment activities in a more organized and systematic way.

1.4 INITIAL FUNCTIONAL REQUIREMENTS

For the initial functional requirement of the event-driven Job Requirement and Application processing system that we will focus on the main activities of both candidates and recruiters with the interaction between services. First, a system needs to support the efficient job and candidate management which is the recruiters need a reliable way to post and provide all the detail about the job vacancies. Candidates play a role as a simple job to create profile, upload resume and apply the jobs of their capabilities so the system exist because of both parties which employers and job seekers get their require to centralized and structured environment of their jobs to interact with.

Besides, the system needs to handle entire recruitment workflow smoothly for real time which including submission application, shortlisting, interview scheduling and decision making for recruiter to hire the candidates which is appropriate with the jobs. In an event-driven, especially important because each step depend on events being processed quickly and accurately to reduce delays and ensure every action trigger immediately as soon as the system receive the notification.

Furthermore, crucial part of the system needs to provide communication and decision support to the candidates because they expect time updates about application status need to insight into hiring process and performance. This needs a notification system and basic analytics or reporting features to ease the candidate for manually checked on their application submission on their jobs. These ensure that users stay informed and organizations can improve their hiring strategies based on user's data to make sure the system is not just functional, but also responsive, scalable and user-focused to what they need.

1.5 TEAM MEMBER ROLES

Name	Roles
MUHAMAD NUR IMAN HAFIZI BIN MOHD ADIB	Database Lead
MUHAMMAD HARITH RIEZHAN BIN SYANAZ	Backend Lead
MUHAMMAD SYUKRAN BIN MOHD ZAKRI	Frontend Lead

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